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February 20, 1968
SUPERSEDING
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FEDERAL SPECIFICATION

INSULATION, THERMAL, (FLEXIBLE UNICELLUR SHEET AND PIPE COVERING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers unicellular elastomeric thermal insulation for condensation control at temperatures -100° to + 200°F. (-73.3° to + 93.3°C.).

1.2 Classification. The thermal insulation covered by this specification shall be of the following classes, as specified (see 6.2):

Class T - Tubular.
Class S - Sheet.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activites outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

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Military Specification:

MIL-L-10547 - Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Publications:

C 177 - Test for Thermal Conductivity of Materials by Means of the Gaurded Hot Plate.
C 355 - Test for Water Vapor Transmission.
D 1056 - Specifications and Methods of Test for Sponge and Expanded Cellular Rubber Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Classification Board:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Boulevard, Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. The material shall be unicellular composition, such as expanded synthetic elastomer, suitable for use as thermal insulation and condensation control.

3.2 Class T, tubular (see 6.2).

3.2.1 Inside diameters and tolerances. The inside diameter shall be within the following tolerances:

Inside diameter Inches	Tolerances Plus, inches - Minus, nothing
Up to 1/2	1/16
1/2 to 1	1/8
1 to 1-5/8	3/16
1-5/8 to 2-3/8	1/4
2-3/8 and over	3/8

3.2.2 Wall thickness. The wall thickness shall be as specified (see 6.2), and shall be within the following tolerances:

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Wall thickness Inches	Tolerance Plus, inches - Minus, nothing
3/4	1/8
3/4 and over	3/16

3.2.3 Length. The length of tubular insulation shall be as specified (see 6.2), with a tolerance of plus 3 percent or minus 1 percent. A maximum of 10 percent short lengths each with a minimum length of 18 inches, is permissible.

3.3 Class S, sheet (see 6.2).

3.3.1 Thickness. The thickness of sheet shall be within the following tolerances:

Thickness Inches	Tolerance Plus, inch	Minus, inch
Up to 1/2	1/16	1/16
1/2 and over	3/32	3/32

3.3.2 Length and width. The length and width of sheet shall be within the following tolerances:

Length or width Inches	Tolerances Plus, inch	Minus, inch
Up to 6	1/4	1/4
6 to 12	3/8	3/8
12 and over	3 percent	3 percent

3.4 Classes S and T, sheet and tubular.

3.4.1 Density. The density shall be not less than 4 or more than 8.5 p.c.f. (see 4.4.2).

3.4.2 Water absorption. The water absorption shall not be greater than 10 percent by weight (see 4.4.3).

3.4.3 Flexibility. When bent 180° in accordance with 4.4.4, insulation shall not crack or split (see 4.4.4).

3.4.4 Thermal conductivity. The thermal conductivity (k) determined at a mean temperature of 75°F. shall not be greater than 0.30 Btu. in/hr. sq. ft. deg F. (see 4.4.5).

3.4.5 Water vapor transmission. The permeability shall not be greater than 0.30 perm. inch (see 4.4.6).

3.5 Workmanship. Thermal insulation shall be free of holes, dirt, and extraneous particles.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all insulation of the same class and size submitted for inspection at the same time.

4.2.2 Visual examination of end item. Sampling shall be in accordance with level S-3 in MIL-STD-105.

4.2.3 Examination of packing. Sampling shall be in accordance with level S-4.

4.2.4 Samples for tests. Sampling shall be in accordance with level S-1.

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4.3 Examination.

4.3.1 End item. Insulation shall be examined for defects listed in table I. The acceptable quality levels (AQL's) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Class not as specified	X	
Dimensions not as specified, or not within tolerance	X	
Holes punched in insulation	X	
Dirt present		X
Extraneous particles present	X	
Damage or defects not affecting function or serviceability		X
Damage or defects affection function or serviceability	X	

4.3.2 Examination of preparation for delivery. An examination shall be made to determine that packing and marking requirements of the applicable contract or order are complied with. Defects shall be scored in accordance with table II. The lot size shall be the number of shipping containers fully prepared for delivery with the exceptions that containers need not be sealed or closed. Examination shall be made in two phases. First an interior examination in process of packing, and second an examination of containers fully prepared for delivery. The sample unit for each of the two phases shall be one container prepared for delivery as set forth above. The inspection level shall be S-2 of MIL-STD-105 with an AQL of 4.0 percent defective.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Marking (exterior container)	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Workmanship (as applicable)	Inadequate or improper packing such as closure of container flaps, taping of seams, corners, and manufacturer's joint, loose strapping or tape banding; inadequate stapling, bulging, or distortion of container.

4.4 Test methods. All tests shall be made on samples conditioned at 73.4° ± 3.6°F. (23° ± 2°C.) for 24 hours in an atmosphere of 50 ± 2 percent relative humidity. AQL for tests shall be 4.0 percent defective.

4.4.1 Measurement of specimens.

4.4.1.1 Class T, tubular. The length and wall thickness shall be measured with a steel rule graduated in 0.01 inch increments. Care should be taken not to distort the tube. At least three measurements shall be taken distributed around the circumference. The inside diameter shall be measured with a plug gage graduated in 1/32 inch increments.

4.4.1.2 Class S, sheet. The length and width shall be measured with a steel rule graduated in 0.01 inch increments. Care should be taken not to distort the sheet. The thickness up to and including two inches shall be measured with a micrometer gage with a presser foot of 1-1/4 inch diameter applied to the sample with a 25 gram force. The other flat bearing surface shall not be smaller than the presser foot. The scale shall be graduated in 0.001 inch increments. Thickness over two inches shall be measured with a steel rule graduated in 0.01 inch increments. At least three measurements shall be taken.

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4.4.2 Density (see 3.4.1).

4.4.2.1 Class T, tubular. The size of the specimens shall be 4 to 8 inches in length. Weigh each specimen with a balance sensitive to 0.01 grams. The volume shall be obtained from the average measurements of the length, inside diameter and the wall thickness.

4.4.2.2 Class S, sheet. The size of the specimens shall be 3 by 4 inches by specified thickness. Weigh each specimen with a balance sensitive to 0.01 gram. The volume shall be obtained from the average measurements of the length, width, and thickness.

4.4.3 Water absorption. Except as noted below, water absorption shall be tested in accordance with ASTM D 1056 (see 3.4.2).

4.4.3.1 Class T, tubular. The specimens shall be a two inch length of the tubular material, with the specified inside diameters and wall thicknesses. The volume of a specimen shall be determined from the average measurements of the length, inside diameter and wall thickness.

4.4.3.2 Class S, sheet. The specimens shall be discs having a 4 square inch area x the specified thickness. The volume of a specimen shall be determined from the average measurements of the diameter and thickness.

4.4.4 Flexibility (see 3.4.3).

4.4.4.1 Class T, tubular. The specimens shall be of size and wall thickness specified, and of suitable length. Across the length rapidly bend the specimen by hand 180° around a mandrel having a diameter not greater than the specimen thickness.

4.4.4.2 Class S, sheet. The size of the specimens shall be 6 inches x 3 inches x specified thickness. Across the 6 inch dimension rapidly bend the specimen by hand 180° around a mandrel having a diameter not greater than the specimen thickness.

4.4.5 Thermal conductivity. The k factor shall be determined in accordance with ASTM C 177 (see 3.4.4).

4.4.6 Water vapor transmission (see 3.4.5).

4.4.6.1 Method. The water vapor permeability tests shall be performed according to the water method in ASTM C 355.

4.4.6.2 Attachment of specimens. The specimens shall be attached to the test dish so that the skin, if any is toward the higher humidity side.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Sheets and pipe covering of like description shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, type III, PPP-B-621, class 2, PPP-B-636, class weather-resistant, or PPP-B-640, class 2, grade A. The wood boxes shall be closed and strapped in accordance with the applicable specification or appendix thereto. The fiberboard boxes shall be closed, waterproof sealed and strapped in accordance with the appendix thereto.

5.1.2 Level B. Sheets and pipe covering of like description shall be packed in accordance with 5.1.1 except the boxes shall be the domestic type or class and the case liner or waterproofing requirements are not required.

5.1.3 Level C. Sheets and pipe covering shall be packed to insure carrier acceptance and safe delivery to destination. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

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5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by contract or order, shipping containers shall be marked in accordance with Fed. Std.No. 123.

5.2.2 Military agencies. In addition to markings required by contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. These flexible insulations are used to prevent excessive heat gain or loss and to prevent condensation of liquid cooling, chilled water and other piping and ducts, tanks, vessels, and other types of equipment.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class (see 1.2).
(c) Clas T, tubular-inside diameter, wall thickness, and length (see 3.2).
(d) Class S, sheet-thickness, length and width (see 3.3).
(e) Selection of applicable level of packing required (see 5.1).
(f) Marking required (see 5.2).

MILITARY INTEREST:

Custodians:

Army - ME
Navy - YD
Air Force - 84

Review activities:

Army - ME, CE
Navy - YD, SH
Air Force - 84

User activities:

Navy - AS, CG

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 5 cents each.

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